

Lecture 01 - Evolution of Computer

Overview

Evolution of Computers, Counting Methods, Number Systems, Binary, Decimal, Octal, Hexadecimal, and Number System Conversions.

What is a Computer?

Computer is derived from the word 'Compute', meaning 'to calculate'. It accepts input, processes data, produces output, and stores information.

Why Computers Were Invented

Manual calculations were slow, error-prone, and time-consuming. Computers were developed to perform calculations quickly and accurately.

Early Counting Methods

1. Finger Counting
2. Tally Marks (used for attendance, trade, inventory, animal counting).

Number Systems

Decimal (Base 10), Binary (Base 2), Octal (Base 8), Hexadecimal (Base 16).

Decimal Number System

Digits: 0-9. Example: $386 = 3 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$.

Binary Number System

Digits: 0 and 1. 0 = OFF, 1 = ON. Computers understand binary because transistors have ON/OFF states.

Transistor

A transistor has two states: ON (1) and OFF (0). It is the basic building block of modern computers.

Decimal to Binary

Repeatedly divide the decimal number by 2 and read remainders from bottom to top.
Example: $27 = 11011_2$.

Binary to Decimal

Multiply each bit by 2 raised to its position. Example: $1101_2 = 13_{10}$.

Octal Number System

Base 8. Digits: 0-7. Example: $45_8 = 37_{10}$.

Hexadecimal Number System

Base 16. Digits: 0-9, A-F (A=10 ... F=15). Example: $26_{16} = 38_{10}$.

Important Formulas

Decimal→Binary: Divide by 2.

Binary→Decimal: $\text{Digit} \times 2^{\text{Position}}$.

Octal→Decimal: $\text{Digit} \times 8^{\text{Position}}$.

Hex→Decimal: $\text{Digit} \times 16^{\text{Position}}$.

Important Viva Questions

What is a Computer? Why Binary? What is a Number System? Explain Decimal, Binary, Octal, Hexadecimal. Perform conversions. What is a Transistor?

Summary

Computer means 'to compute'. Binary is the language of computers. Decimal uses Base 10, Binary Base 2, Octal Base 8, Hexadecimal Base 16.